

t -distribution

ECON 97: Economic Toolkit

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<https://join.iclicker.com/RDNG>

Confidence Interval Derivation

For a 95% confidence interval:

$$P(-1.96 < Z < 1.96) \approx 0.95 \quad (z_{\alpha/2} = 1.96, \alpha = 0.05)$$

Substitute the standardized sample mean:

$$P\left(-1.96 < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < 1.96\right) \approx 0.95$$

Rearrange:

$$P\left(\bar{X} - 1.96\frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + 1.96\frac{\sigma}{\sqrt{n}}\right) \approx 0.95$$

Confidence Interval Formula

If we know the true variance, use σ

If we don't know σ , use sample standard deviation s

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$\bar{X} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$$

where $z_{\alpha/2}$ is the “critical value” at the α level. Since this is a 2-sided interval, the α probability is split between two sides of the distribution

$$\alpha = 0.1 \implies z_{\alpha/2} = z_{0.05} = 1.645$$

$$\alpha = 0.05 \implies z_{\alpha/2} = z_{0.025} = 1.96$$

$$\alpha = 0.01 \implies z_{\alpha/2} = z_{0.005} = 2.576$$

CLT + Confidence Interval practice

A sample of $n = 64$ observations has

$$\bar{X} = 82, \quad s = 12.$$

Using the large-sample approximation, a 95% confidence interval is

$$\bar{X} \pm 1.96 \frac{s}{\sqrt{n}}$$

$$82 \pm 1.96 \left(\frac{12}{\sqrt{64}} \right) = 82 \pm 2.94$$

$$(79.06, 84.94)$$

The Student's t distribution

If n is small (few observations) and σ is unknown, the standard normal distribution might not give a good approximation

If σ were known, we could still use

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1).$$

For confidence intervals in small samples, use **Student's t distribution**

$$T = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t(n - 1).$$

Properties of t distribution

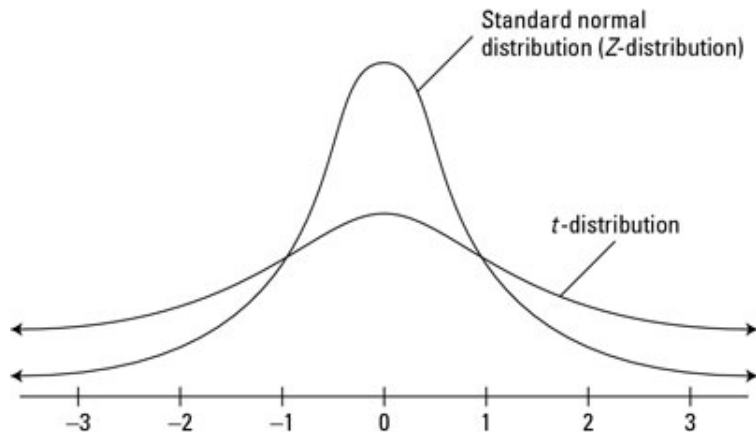
The t distribution looks like the standard normal distribution:

- centered at 0
- symmetric
- bell-shaped

But it has **fatter tails** (larger critical values), so confidence intervals are wider

As the sample size n grows, the t distribution approaches the standard normal distribution

Distribution of t vs. Z



Degrees of freedom

For a sample of size n , the relevant t distribution has

$$\text{degrees of freedom} = n - 1$$

Why $n - 1$?

- We estimate μ using $\bar{X}$
- Once $\bar{X}$ is fixed, only $n - 1$ deviations can vary freely
- This is also why sample variance divides by $n - 1$

Small-sample confidence interval

A $100(1 - \alpha)\%$ confidence interval for μ is

$$\bar{X} \pm t_{\alpha/2}(n - 1) \frac{s}{\sqrt{n}}$$

For a 95% confidence interval:

$$\bar{X} \pm t_{0.025}(n - 1) \frac{s}{\sqrt{n}}$$

Interpretation: the interval is centered at $\bar{X}$, and the margin of error is

$$t_{\alpha/2}(n - 1) \frac{s}{\sqrt{n}}$$

Choosing the critical value

The critical value depends on two things:

- the confidence level
- the degrees of freedom, $n - 1$

For a 95% confidence interval, use $\alpha = 0.05$:

$$\text{critical value} = t_{0.025}(n - 1)$$

For a 90% confidence interval, use $\alpha = 0.10$:

$$\text{critical value} = t_{0.05}(n - 1)$$

Example: 90% confidence interval

Suppose a random sample of size $n = 20$ from a normal population has

$$\bar{X} = 507.50, \quad s = 89.75$$

For a 90% confidence interval:

$$\alpha = 0.10, \quad df = n - 1 = 19$$

From the t table:

$$t_{0.05}(19) = 1.729$$

Example: 90% confidence interval

$$\bar{X} \pm t_{0.05}(19) \frac{s}{\sqrt{n}}$$

$$507.50 \pm 1.729 \left(\frac{89.75}{\sqrt{20}} \right)$$

$$507.50 \pm 34.70$$

$$(472.80, 542.20)$$

“90% confident that the population mean is between 472.80 and 542.20”

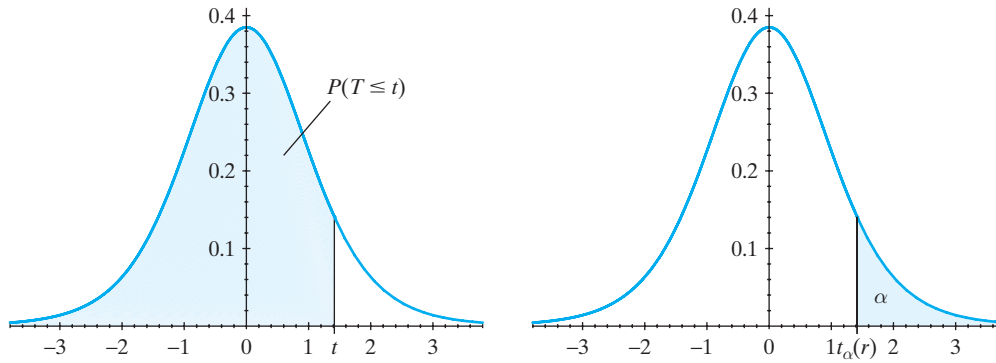
Practice: t confidence interval

A sample of $n = 15$ observations from a normal population has

$$\bar{X} = 42.4, \quad s = 6.8.$$

Find a 95% confidence interval for μ

Table VI The t Distribution



$$P(T \leq t) = \int_{-\infty}^t \frac{\Gamma[(r+1)/2]}{\sqrt{\pi r} \Gamma(r/2) (1 + w^2/r)^{(r+1)/2}} dw$$

$$P(T \leq -t) = 1 - P(T \leq t)$$

	$P(T \leq t)$						
	0.60	0.75	0.90	0.95	0.975	0.99	0.995
r	$t_{0.40}(r)$	$t_{0.25}(r)$	$t_{0.10}(r)$	$t_{0.05}(r)$	$t_{0.025}(r)$	$t_{0.01}(r)$	$t_{0.005}(r)$
1	0.325	1.000	3.078	6.314	12.706	31.821	63.657
2	0.289	0.816	1.886	2.920	4.303	6.965	9.925
3	0.277	0.765	1.638	2.353	3.182	4.541	5.841
4	0.271	0.741	1.533	2.132	2.776	3.747	4.604
5	0.267	0.727	1.476	2.015	2.571	3.365	4.032
6	0.265	0.718	1.440	1.943	2.447	3.143	3.707
7	0.263	0.711	1.415	1.895	2.365	2.998	3.499
8	0.262	0.706	1.397	1.860	2.306	2.896	3.355
9	0.261	0.703	1.383	1.833	2.262	2.821	3.250
10	0.260	0.700	1.372	1.812	2.228	2.764	3.169
11	0.260	0.697	1.363	1.796	2.201	2.718	3.106
12	0.259	0.695	1.356	1.782	2.179	2.681	3.055
13	0.259	0.694	1.350	1.771	2.160	2.650	3.012
14	0.258	0.692	1.345	1.761	2.145	2.624	2.997
15	0.258	0.691	1.341	1.753	2.131	2.602	2.947
16	0.258	0.690	1.337	1.746	2.120	2.583	2.921
17	0.257	0.689	1.333	1.740	2.110	2.567	2.898
18	0.257	0.688	1.330	1.734	2.101	2.552	2.878
19	0.257	0.688	1.328	1.729	2.093	2.539	2.861
20	0.257	0.687	1.325	1.725	2.086	2.528	2.845
21	0.257	0.686	1.323	1.721	2.080	2.518	2.831
22	0.256	0.686	1.321	1.717	2.074	2.508	2.819
23	0.256	0.685	1.319	1.714	2.069	2.500	2.807
24	0.256	0.685	1.318	1.711	2.064	2.492	2.797
25	0.256	0.684	1.316	1.708	2.060	2.485	2.787
26	0.256	0.684	1.315	1.706	2.056	2.479	2.779
27	0.256	0.684	1.314	1.703	2.052	2.473	2.771
28	0.256	0.683	1.313	1.701	2.048	2.467	2.763
29	0.256	0.683	1.311	1.699	2.045	2.462	2.756
30	0.256	0.683	1.310	1.697	2.042	2.457	2.750
∞	0.253	0.674	1.282	1.645	1.960	2.326	2.576

This table is taken from Table III of Fisher and Yates: *Statistical Tables for Biological, Agricultural, and Medical Research*, published by Longman Group Ltd., London (previously published by Oliver and Boyd, Edinburgh).